

## 2017-Ph-H1-Q11

**Section:** Our Dynamic Universe

**Topic:** Forces, Energy and Power

A block is pushed along a surface. Work done against friction is asked.

Given:

- Force = 20 N
- Distance = 3 m
- Frictional force = 4.0 N

Work done **against friction** =  $F_{\text{friction}} \times d$

$$W = 4.0 \times 3 = 12 \text{ J}$$

 **Final Answer: C — 12 J**

 **Revision Tips:**

- Work = Force × distance
- Only include the force **in the direction of motion**